Kaleb Rice

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Education

Bachelor of Science in Mechanical Engineering – Portland State University — September 2022 – Ongoing

GPA: 3.73 **Expected Graduation:** June 2027

Relevant Courses: Finite Element Analysis (FEA) · Materials Failure Analysis · Heat Transfer and Thermodynamics · Fluid Mechanics · Vibrations & System Dynamics · Manufacturing Processes

Technical Skills

CAD & Design: Creo · SolidWorks (CSWA) · OnShape (COP) · Fusion 360 · Siemens Solid Edge · FreeCAD · KiCad

Fabrication: FDM & SLA 3D Printing · MasterCAM · Mill · Lathe · MIG / TIG Welding

Software & Coding: MATLAB · C++ · Python · Arduino / ESP32 Microcontrollers

Additional: Design for Manufacturing (DFM) · Geometric Dimensioning and Tolerancing (GD&T)

Experience

Automation Engineering Intern, Epson Portland Inc. — April 2026 – September 2026

- Designed a modular adhesive applicator that reduced production downtime by 20 minutes each day.
- Managed the design and ordering of a cleanroom conveyor line to enable automatic packaging of flexible ink bags.
- Worked across departments to collect data and effectively implement various assembly line modifications to improve worker safety, efficiency, and quality of life.

Machine Shop Technician, PSU Mechanical Engineering Lab Team (MELT) — October 2025 – May 2026

- Trained students of various skill levels on safe operation of mills, lathes, welders, and other shop tools.
- Designed and fabricated equipment for raw material processing to improve production efficiency.

Mechanical Design Engineering Intern (MECOP), Benchmade Knife Company — April 2025 – September 2025

- Designed 11 production-ready knife assemblies in Creo from concept to manufacturing, integrating DFM and material analysis principles.
- Researched and prototyped new compliant mechanisms, finishes, and manufacturing techniques to improve reliability and aesthetics.
- Used GD&T fundamentals to create detailed engineering drawings and tolerance analyses for production use.

Lead Mechanical Engineer, Portland State Aerospace Society — February 2023 – March 2025

- Directed mechanical design for a $200k open-source satellite communications station (UniClOGS), collaborating with multidisciplinary teams.
- Modeled structural components and performed CFD-based wind-load simulations to validate city permitting calculations.
- Trained new colleagues in CAD best practices, documentation, and design reviews to ensure repeatable, manufacturable designs.

Freelance Engineering, Kale Creative Works — December 2021 – Ongoing

- Provided nationwide CAD, 3D printing, CNC programming, and prototyping services for individuals and small businesses.
- Managed full client lifecycle: concept development, design consultation, manufacturing coordination, and assembly.

Projects & Leadership

Open-Source Plastic Scanner — October 2024 – March 2025

- Awarded $1,000 PSU Cleantech Challenge grant to prototype a near-infrared spectroscopy system for plastic recycling and filament production.

High School Makerspace Founder — May 2022 – Ongoing

- Established and led a community makerspace offering STEAM education and fabrication tools; annual exhibitor at the Portland Winter Light Festival.

FIRST Robotics — September 2018 - February 2020

- Led the mechanical design of an omnidirectional drive robot and oversaw competition strategy.
- Returned as a volunteer judge in 2024 to supervise competitions and interview student teams.